

ASSIGNMENT – 4.5

1. A user reports their laptop won't power on even when plugged in. List 5 troubleshooting steps you would take to diagnose the cause, and for each step, explain what you are checking for.

1) Check the power source and adapter connection

What you're checking: Whether the wall outlet, power strip, and AC adapter are actually delivering power.

How: Plug another device (lamp/phone charger) into the same outlet; ensure the adapter is firmly connected at both ends (wall and laptop).

2) Inspect the charger, cable, and laptop power port

What you're checking: Physical damage that can block power (frayed cable, bent pins, loose port).

How: Look closely at the adapter LED (if present), cable insulation, connector tip, and the laptop's DC-in/USB-C port for looseness or debris.

3) Perform a hard reset (power drain)

What you're checking: Whether residual charge or a stuck power state is preventing boot.

How: Unplug charger, remove battery if removable, hold power button 30–60 seconds, then reconnect charger (and battery) and try powering on.

4) Test with a known-good compatible charger / different port

What you're checking: If the original adapter or specific charging port is faulty.

How: Use another compatible charger or a different USB-C power port (if supported) and see if the laptop powers on or shows a charging LED.

5) Check battery and power indicators / try without battery

What you're checking: Whether the battery is dead/failed or the system only runs on AC.

How: Observe charging LEDs; if battery is removable, take it out and try powering on with AC only. If it starts, the battery likely needs replacement.

2. You receive a helpdesk ticket: 'My monitor is showing a blank screen, but the CPU seems to be running.' Write a step-by-step troubleshooting flow you would follow to identify and resolve the issue.
Hint: Consider checking both hardware connections and basic display settings.

1) Confirm the PC is actually on: Listen for fans/HDD, check power LED, keyboard caps-lock LED. If no signs of life, treat as a no-power issue instead.

2) Check monitor power and input source: Ensure monitor power cable is plugged in and the monitor is ON. Use the monitor's OSD buttons to select the correct input (HDMI/DP/VGA) matching the cable you used.

3) Reseat/replace video cable and try another port: Disconnect and firmly reconnect both ends of the video cable. If possible, swap the cable or try a different video port on the GPU/motherboard.

4) Test with another display (or another cable): Connect the PC to a different monitor/TV, or connect the suspect monitor to a different PC/laptop. This isolates whether the fault is the monitor, cable, or PC video output.

5) Check basic display settings and GPU detection: If you can get any image (e.g., BIOS/another monitor), ensure the OS isn't outputting to a different display (Windows: Win+P → PC screen only/Duplicate). In Device Manager, verify the display adapter is present and not disabled; update/reinstall GPU driver if needed.

6) Internal hardware checks (if still blank): Reseat RAM and GPU (if desktop), clear CMOS, and listen for beep codes/POST. A blank screen with fans spinning often points to RAM/GPU/motherboard POST failure.

3. A user complains that their wireless mouse is unresponsive, but the keyboard works fine. Describe how you would isolate whether the issue is with the mouse, the USB port, or the system settings.

1) Basic mouse checks

Replace/charge the mouse batteries; ensure the mouse is switched ON. Re-pair/reconnect: for 2.4 GHz dongle, unplug and replug; for Bluetooth, remove device and pair again.

2) Test the mouse on another computer (or test another mouse on this PC)

If the same mouse fails on another PC → mouse hardware/receiver is likely faulty. If another mouse works on this PC → original mouse is the problem.

3) Change USB port and remove conflicting USB devices

Plug the receiver into a different USB port (preferably direct on PC, not a hub). Shut down, disconnect non-essential USB devices, boot, and test the mouse to rule out USB conflicts.

4) Check system settings and drivers

In Device Manager, look under 'Mice and other pointing devices' and 'Universal Serial Bus controllers' for warnings; update/reinstall mouse and USB controller drivers. Ensure the mouse isn't disabled in settings and that pointer/USB selective suspend settings aren't interfering.

- **Mouse faulty:** if it fails on multiple PCs or another mouse works fine on this PC.
- **USB port/receiver issue:** if it works in some ports/PCs but not others, or if removing other USB devices fixes it.
- **System settings/driver issue:** if hardware tests pass but Device Manager shows errors or reinstalling drivers resolves it.

4. Imagine you are working on a helpdesk team for a gaming cafe. One system is overheating and shutting down during use. List 3 possible hardware causes and describe how you would check each one.
Hint: Think about fans, dust, and thermal paste.

1) Dust-clogged fans/heatsinks and poor airflow

Check: Power off, open the case, visually inspect CPU/GPU fans and radiators for dust; check case intake/exhaust fans and cable blockage.

Fix: Clean with compressed air, ensure fans spin freely, and improve cable management/airflow.

2) Failing or stalled CPU/GPU fans

Check: With the side panel off (carefully), power on and watch whether CPU and GPU fans spin up and accelerate under load; listen for grinding/noise.

Fix: Replace non-spinning or noisy fans; reseat fan power headers on the motherboard.

3) Dried/insufficient thermal paste or poor cooler contact

Check: If temperatures spike quickly under load despite clean fans, suspect thermal interface issues. Remove the cooler, clean old paste, inspect for uneven contact or bent mounting.

Fix: Reapply quality thermal paste correctly and re-mount the cooler with even pressure; verify temperatures drop under load.